

StudyMate: A Developer Collaboration Platform

Group: Web Crew

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Brief about the Project

Problem: Students and developers often struggle to find dedicated coding partners with similar skill sets and goals for projects.

Solution: *StudyMate* is a streamlined social platform designed exclusively for developers.

Core Features:

- **Find Partner:** An algorithm to discover and connect with like-minded developers based on tech stacks.
- **Community Posts:** A dedicated feed to share project updates, ask technical questions, and showcase work.
- **Developer Profiles:** Customizable profiles highlighting individual skills, GitHub links, and current focuses.

Comparison with Existing Systems

Feature	General Social Media	StudyMate (Our Project)
Primary Focus	Broad networking, job hunting, or general discussions.	Strict focus on finding active project/coding partners.
Matchmaking	Manual searching and cold messaging.	Tech-stack specific filtering and matching.
Distractions	High (ads, viral posts, recruiters).	Low (strictly developer and project-centric content).

Tools Being Used

This project is being developed using the **MERN Stack** for a robust, scalable, and responsive application.

- **Frontend:** React.js (built with Vite), Tailwind CSS for rapid UI styling.
- **Backend:** Node.js with Express.js for handling API routes and business logic.
- **Database:** MongoDB (NoSQL) for flexible data storage of user profiles and posts.
- **Authentication:** JSON Web Tokens (JWT) for secure user sessions.

Timeline of Project (6 Weeks)

- **Week 1:** Requirement analysis, UI/UX wireframing, and repository setup.
- **Week 2:** Backend architecture, database schema design, and implementing User Authentication.
- **Week 3:** Developing the Frontend layout and integrating the Profile Page.
- **Week 4:** Building the Community Posts feature (Create, Read, integration with backend).
- **Week 5:** Implementing the "Find Partner" filtering logic and UI.
- **Week 6:** Final integration, bug fixing, responsive design tweaks, and project documentation.

Project Status & Deployment

Completed Milestones:

- ✓ Frontend (React/Vite) and Backend (Node/Express) fully integrated.
- ✓ User Authentication, Profile Pages, Post Section, Find Partner and Community

Project Deployment:

- **Current Status: Fully Completed and Hosted**
- **Infrastructure:** Microsoft Azure Cloud
- **Live Application:** <http://4.186.31.157/>

Thank You!